

Ethical Issues in Biomedical Research: Historical Legacies, Gender Inclusivity, and Genomic Frontiers

Biomedical research has come a long way, not just because science has advanced, but also because we've had to face some really difficult ethical issues along the way. From horrific experiments during **World War II** to today's debates about gene editing, researchers have constantly had to ask: **How far is too far? And how do we protect people while still making progress?**

In their article, "**Ethical Issues in Research**" (2017), **Raul Artal and Sheldon Rubenfeld** talk about how ethics in medical research have evolved. Building on what they wrote, I want to talk about **three big topics**:

1. What we learned from the past
2. How we're working toward more inclusive research
3. The ethical challenges of powerful new technologies like gene editing.

What History Taught Us About Ethics in Research

During the **Nazi regime**, doctors carried out horrifying experiments on people—mostly on those already being targeted: **Jews, Roma, people with disabilities, and others**. They claimed it was for science or the good of the state, but it was cruelty disguised as research. **Artal and Rubenfeld** explain how this twisted thinking—this idea that the ends justified the means—led to some of the worst abuses in medical history.

After the war, the world responded by creating the **Nuremberg Code**, a set of rules to make sure research would never again be done that way. It said things like: participants must volunteer, they must give informed consent, and they must be allowed to leave a study at any time. Later on, other guidelines like the **Declaration of Helsinki** added more protections.

Today, before any study involving people can move forward, it has to go through an ethics board (called an **IRB**). The point is to make sure people's rights, safety, and dignity are respected. So while we've made big improvements, the hard lessons from history are still shaping how we do things—and that's a good thing.

Why Inclusion in Research Matters—and Why It's Tricky

For a long time, women—especially those who could get pregnant—were left out of medical research. People were worried that if a woman got pregnant during a study, it might harm the fetus. While that concern wasn't totally wrong, leaving women out completely created another big problem: most of our medical knowledge came from studying men.

That doesn't work. Women's bodies are different—hormones, metabolism, how diseases show up—it all matters. By excluding them, science ended up being unfair and incomplete.

Artal and Rubenfeld point out that not only was this exclusion unfair, it was bad science. You can't apply results from men to women and expect the same outcomes. Right now, there's a bigger emphasis on including women and other groups that are usually not part of research. This makes studies more thorough and fair.

But adding pregnant women to research brings up new ethical questions: How can we protect the baby while still respecting the woman's right to make decisions about her own body?

It's hard to find the right balance. That's why research involving pregnant women needs more attention, clear communication, and careful review. There isn't an easy answer yet—but ignoring the issue is not an option anymore.

The Ethical Risks Associated with Gene Editing

Emerging technologies such as **CRISPR** are remarkably potent—they enable scientists to modify segments of an individual's DNA. This advancement holds the potential to eradicate genetic disorders, prevent cancer from developing, or even avert hereditary conditions.

However, this ability raises important ethical concerns. **Artal and Rubenfeld** say that if gene editing isn't properly watched, it could be used in bad ways—such as trying to make '**perfect**' kids by picking traits like smartness, appearance, or sports skills. That's not just sci-fi anymore; it's something we're going to have to deal with in real life. And if only rich people can afford those kinds of treatments, we could end up making social inequality even worse.

There are also big questions about consent. If we change a baby's genes before they're born, they don't get to say yes or no. **What if we make a mistake? What if there are side effects later on?** These aren't simple "yes or no" questions—they're the kind that need long-term thinking, careful oversight, and global conversations.

If we rush things, we could end up causing harm. On the other hand, if we take our time, we might delay treatments that could actually save lives. The objective we face is to find a way that honors people's rights and ensures that no one is left out.

Bringing It All Together

In the field of research ethics, it's not just about following rules—it's about ensuring that science maintains its human element. Whether we are reflecting on previous errors, choosing who to involve in our research, or figuring out how to utilize powerful new technologies, **we must keep asking ourselves: Is this fair? Is this the right direction to go?**

Artal and Rubenfeld stress the need to stick to basic values: treating people with respect, promoting good, avoiding harm, and ensuring fairness for everyone.

Even though the world is changing fast, these values are still very important. By focusing on them in our learning, we can achieve great things while keeping our humanity.

Reference

Artal, R., & Rubinfeld, S. (2017). Ethical issues in research. *Best Practice & Research Clinical Obstetrics & Gynaecology*, 43, 107–114.
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